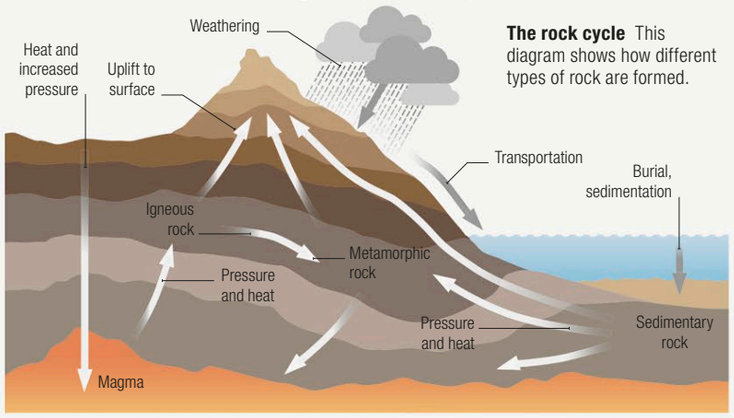
Gems in the Earth

The origins of rocks, minerals, and gemstones

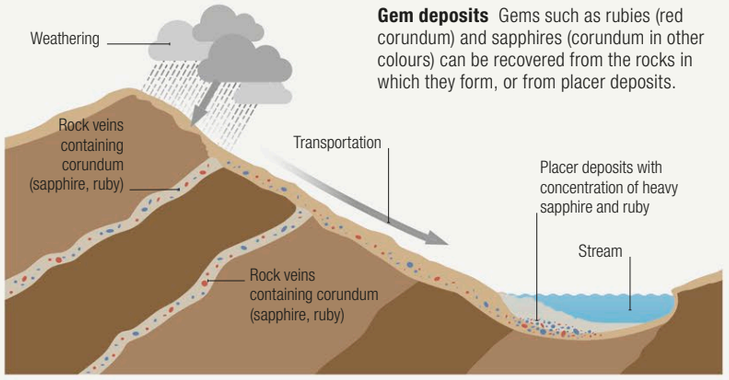
Rocks and minerals are created in the rock cycle. All rocks begin as igneous, but over time they are altered by re-melting, erosion, or metamorphosis – weathering and erosion lead to the formation of sedimentary rock, which can turn into metamorphic rock through temperature or pressure conditions.



The rock cycle This diagram shows how different types of rock are formed.

GEMS IN NATURE

Mineral gemstones are directly mined from the rocks in which they were originally formed (see p.25): examples of these include diamond, tanzanite, ruby, kyanite, celestine, emerald, tourmaline, and aquamarine. Other types of gem, released from their original rock by weathering, can be mined from placer deposits found in stream gravels. Examples of these varieties include topaz, sapphire, chrysoberyl, garnet, zircon, and spinel.



Gem deposits Gems such as rubies (red corundum) and sapphires (corundum in other colours) can be recovered from the rocks in which they form, or from placer deposits.





Crystals

A crystal is a solid, the component atoms of which are arranged in a particular, repeating, three-dimensional pattern. When these internal patterns produce a series of external, flat faces arranged geometrically, a crystal, such as the rhodochrosite above, is created.



Native elements

These are chemical elements that occur in nature uncombined with other elements (see also p.14), including gold, silver, and diamond (carbon). With the exception of platinum and gold, most metals are extracted from minerals that contain them.



Organic gems

Created through organic processes, organic gems are not commonly crystalline. Jet is a form of coal, derived from plant matter; coral and shell are the secreted skeletons of marine animals, while pearls are formed by shelled molluscs; and amber and copal are tree resins, fossilized and semi-fossilized. Organic gems are generally softer than minerals and easier to work.